

Shadaab Ghani

Stockholm, Sweden

Email: shadaab@kth.se

GitHub: github.com/shadaabghani1

ResearchGate: researchgate.net/profile/Shadaab-Ghani

Professional Summary

AI/ML engineer and doctoral researcher with 5+ years of industry experience building production-ready AI and machine learning systems, along with hands-on research and development experience in academia. Skilled in generative AI, LLM applications, Retrieval-Augmented Generation (RAG), agentic AI systems, Python backend development, APIs, NLP, and applied machine learning. Experienced in taking systems from concept to deployment, including architecture, implementation, evaluation, and user-facing delivery.

Skills

Programming: Python, SQL, Rust, R, C++, JavaScript

Generative AI and LLMs: LLM applications, RAG, agentic AI systems, LangGraph, prompt engineering, hallucination detection, source attribution

Machine Learning and NLP: PyTorch, TensorFlow, scikit-learn, CatBoost, NLI, sentiment analysis, information extraction, knowledge graphs, conversational AI

Backend and APIs: FastAPI, REST APIs, WebSocket services, Python backend development

Data and Databases: Pandas, SQL, SQL Server, MongoDB, Redis

Cloud and Tools: Azure, AWS SageMaker, Git, Hadoop, Power BI, MATLAB

Web: React

Experience

KTH Royal Institute of Technology, Stockholm, Sweden

Doctoral Student, Software and Computing Systems

Aug 2024 – Present

- Researching hallucination detection in Retrieval-Augmented Generation (RAG), focusing on cases where models' internal knowledge overrides retrieved documents.
- Working on source attribution and response verifiability under conflicting multi-agent and multi-source settings.
- Independently designed, built, and deployed production-ready generative AI systems at KTH, owning the full lifecycle from architecture and backend development to evaluation and delivery.
- Built an agentic AI application using LangGraph, RAG, Python backend services, APIs, and retrieval pipelines for personalized building energy-efficiency advising.
- Published two peer-reviewed conference papers on hallucination detection and conversational agents for building energy efficiency.

ZS Associates, Bangalore, India

Machine Learning Engineer

Jul 2021 – Jun 2024

- Built production-ready AI/ML systems in India, translating business problems into deployable solutions used in real client and internal workflows.
- Led AI-driven product innovation for consumer goods using multimodal LLMs including GPT-4V and LLaVA, and fine-tuned Stable Diffusion for packaging design and concept generation.

- Built an LLM-based sales training platform using GPT-3.5 to simulate customer conversations and automate performance grading.
- Developed NLP pipelines for large-scale customer review analysis including sentiment extraction, phrase-importance modeling, and knowledge graph construction.
- Built predictive sales models using CatBoost and Random Forest, achieving approximately $0.7 R^2$ to inform product concepts and business decisions.

PricewaterhouseCoopers Pvt. Ltd., Kolkata, India

Technical Consultant, Advisory

Jul 2018 – Jul 2019

- Contributed to software for international travel tracking and guideline generation, including feature development, bug fixing, and codebase improvement.
- Built a chatbot proof of concept using BERT and spaCy to automate parts of country-specific travel guideline generation.

Projects

SPARA: Agentic AI System for Building Energy Efficiency

2024 – Present

- Built and deployed an agentic AI system for Stockholm housing cooperatives using LangGraph and RAG over a curated knowledge base of advisory communications.
- Designed routing and multi-agent workflows to deliver personalized energy-efficiency recommendations and support expert evaluation.
- Developed the system as an end-to-end production-ready application, including backend logic, retrieval workflows, and user-facing interaction flows.
- Demo: <https://test.sparabot.com/>
- Production: <https://sparabot.com/>

Hallucination Detection Metric for NLG

2024 – 2025

- Proposed a metric combining entity-level alignment, semantic consistency using NLI, and normalized perplexity for fluency evaluation.
- Evaluated the method on benchmark datasets including BEGIN and FEVER.

Multimodal Concept and Packaging Generation (Industry)

2022 – 2024

- Developed multimodal pipelines for product concept generation and packaging design to support innovation workflows.

Education

KTH Royal Institute of Technology, Stockholm, Sweden

PhD in Computer Science

Aug 2024 – Present

Research: Trustworthy LLMs, RAG evaluation, hallucination detection, source attribution

Chennai Mathematical Institute (CMI), Siruseri, India

M.Sc. in Data Science

Aug 2019 – Jun 2021

CGPA: 8.35/10

Jadavpur University, Kolkata, India

B.E. in Power Engineering

Aug 2014 – May 2018

Overall GPA: 7.98/10

Publications

Towards Trustworthy Conversational Agents: A New Method for Hallucination Detection in NLG

Shadaab Ghani, Anne Håkansson, Oleksii Pasichnyi

Procedia Computer Science, 2025

DOI: 10.1016/j.procs.2025.09.651

Conversational Agents for Building Energy Efficiency – Advising Housing Cooperatives in Stockholm on Reducing Energy Consumption

Shadaab Ghani, Anne Håkansson, Oleksii Pasichnyi, Hossein Shahrokni

IOP Conference Series: Earth and Environmental Science, 2025

DOI: 10.1088/1755-1315/1546/1/012027

Awards

Most Innovative Project Award, ZS Associates

Recognized for innovation in applied AI and generative AI work for consumer goods product development.

Credit Suisse Top Quant Challenge, Chennai Mathematical Institute

Selected among three teams from CMI.

Additional Information

Languages: English (fluent)

Interests: Trustworthy AI, evaluation, RAG systems, sustainability, building energy efficiency